Antonio B. Garcia Jr.

AI / Machine Learning Engineer | Project Lead

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PROFILE

Third-year B.S. Computer Science student on the Machine Learning track at National University Manila. I build ML and LLM systems end to end, from a written specification and clean evaluation to a deployed app, and I lead the teams that ship them. Recent work includes a pre-registered study showing that 99% kidney-disease accuracy comes from leaked lab features, a 0.94 ROC-AUC credit scorecard on a 2.8M-row synthetic dataset I generated, and RAG and multi-agent apps built under hackathon deadlines. Open to AI/ML internships and roles alongside my degree.

EDUCATION

National University Manila · B.S. Computer Science, specialization in Machine Learning — 2024 – 2028 (expected)

- Third-year standing, A.Y. 2026–2027.
- Relevant coursework: Machine Learning, Advanced Database Systems, Software Engineering, Operating Systems, Automata Theory, Data Structures and Algorithms.

AMA Computer Learning Center, Taguig · Senior High School, ICT (Programming Strand) — 2021 – 2023

- Graduated with High Honors, 4.0 GPA.
- Thesis: "Best Practices in Web Development in Accordance to Developers in Makati City", awarded Best Thesis and Best Presenter.

EXPERIENCE

Eskwelabs Innovation Fellowship · Data Modeling Track, Cohort 10 — 2026 · 3 months · Remote

Python, pandas, NumPy, scikit-learn, statsmodels, YAML config, Jupyter

- Built a config-driven synthetic data generator for collections prioritization in consumer lending: a 5-table relational database of **2.83M rows** (50k borrowers, 50k accounts, 600k monthly delinquency snapshots, 1.79M call logs, 344k payments), reproducible from one config file and seed.
- Wrote the specification before the code (Beta, lognormal and hurdle models; 5×5 Markov transition matrices by risk tier; schema dictionary; literature review), built in real-world messiness on purpose (3% missing values, 1% outliers, 0.5% logging errors), and gated every export behind zero-leakage assertions.
- Planted a hidden propensity-to-pay confounder and showed its effect: a naive regression said each collection call lowered recovery (coefficient −4,227); controlling for credit score, income stability and debt burden nearly halved the bias (−2,257).
- Trained default scorecards on features aggregated across all 5 tables: **0.944 ROC-AUC** (Logistic Regression) and 0.939 (Random Forest) on an 81/19 imbalanced split; Random Forest caught 85% of defaults at 0.97 precision.

Sutherland Global Services · Customer Service Representative & Subject Matter Expert — Makati · 2024

- Resolved 80–100+ customer issues a day at a **95%+** satisfaction rating with a low escalation rate; as volunteer Subject Matter Expert, coached 15+ agents in real time, contributing to a **20%** productivity increase in peak weeks.
- Named Best Agent for the highest quality-assurance scores, and Valedictorian of the Wave 3 training class (top of 30+ trainees).

PROJECTS

Beyond 99%: Diagnostic Leakage in CKD Prediction · Solo, Machine Learning course final — GitHub · App · Oct 2026

Python, scikit-learn, pandas, Streamlit, matplotlib, pytest

- Tested where the 99% accuracy reported for chronic kidney disease prediction comes from. Pre-registered a feature-tier ablation on the UCI dataset (400 patients, 24 features) before touching the data.
- Showed it is leakage, not overfitting: with all features, Logistic Regression, Random Forest and Gradient Boosting reached about 99% (98.8% on the held-out test set); with pre-lab features only, accuracy fell to 0.69–0.76 and F2 matched referring everyone.
- Trained a no-lab screening model on 7,363 US adults (NHANES 2017–2020, 13 features a health worker can collect): it flagged **86%** of CKD cases in unseen patients (210 of 243), and with only 300 tests it found **15 more cases** than a points-based risk rule (95% CI 3.9–25.3).
- Deployed it as a Streamlit app that refers a person for testing or ranks a clinic list by available tests, with a pytest suite, model card and an IEEE-format paper being prepared for publication.

AI-Powered Lead Generation Pipeline · AI & Architecture Lead, 2-person team — GitHub · 2026

Python, FastAPI, Playwright, Supabase (PostgreSQL + pgvector), Sentence-Transformers, Groq Llama 3.1, Gemini 2.0 Flash, n8n, Telegram, Docker Compose

- Built a pipeline that turns one Telegram command into audited, scored leads: it scrapes Google Maps, drops duplicates and chains against a self-updating PostgreSQL blacklist, and audits each lead's Facebook, Instagram and website.
- Designed the RAG scoring: each lead is embedded (all-MiniLM-L6-v2) and matched in pgvector against past accepted and rejected leads, so the model sees why similar businesses were taken or dropped before it decides.
- Wrote the evaluation prompts, a batched Groq filter that drops chains and franchises, and Gemini-to-Groq fallback with rate limiting for the audits; approved leads land in a Telegram review queue. Also owned the architecture and n8n stage design; Charles Cabatian owned backend and infrastructure.

Nura: Filipino Healthcare Access Navigator · Project Lead, InnOlympics 2026, 4-person team GitHub · Demo · Apr 2026

FastAPI, React + TypeScript (Vite PWA), Gemini, Vertex AI embeddings, Supabase pgvector, Firestore, Cloud Run

- Led the team to ship in 24 hours a PWA that tells users, in Filipino, Cebuano, Ilocano, Hiligaynon or English, which nearby facility to visit, which benefit to claim (PhilHealth YAKAP, Malasakit, 4Ps, Senior, PWD) and what to bring.
- Designed the architecture and wrote the FastAPI orchestration that ties together facility search, benefit retrieval (RAG over PhilHealth guides) and the Gemini response; split the data pipeline, RAG layer and frontend across teammates and handled integration and release.
- Put a deterministic emergency check in front of everything: chest pain or trouble breathing skips the session lookup, model call and search and sends the user to 911. The last result opens offline, and no user data is stored on the server.

Bantay-Dagitab: Residential Energy Monitor · Project Manager, System Architect & Backend Lead GitHub · 2026

Advanced Database Systems capstone · ESP8266 (NodeMCU), Arduino C++, Django REST Framework, PostgreSQL, FastAPI, LightGBM, Tesseract OCR, OpenCV, Next.js, Docker Compose

- Built and integrated a system that warns post-paid MERALCO households of bill shock before the cycle closes: CT-sensor readings feed a Django/PostgreSQL API, a FastAPI LightGBM service flags usage spikes, OCR digitizes bills and a chatbot explains them.
- Wrote the four JSON Schema contracts (IoT, OCR, anomaly alert, chatbot) that every module had to follow, so the IoT, backend, ML and frontend sub-teams could build in parallel; wrote the largest share of the backend code.
- Owned the IoT side from firmware to database: a clip-on SCT-013 sensor on an ESP8266, with firmware staged as wiring check (bias midpoint at ~1.65 V), calibration, then production, so no reading is trusted before the hardware is verified.
- Designed for the Data Privacy Act (RA 10173): cascade deletes on every user-owned table, JWT refresh-token rotation with blacklisting, and a chatbot that falls back cleanly when no LLM key is set.

JuryBee: Multi-Agent Contract Analyzer · Team Lead (team Joybread Studios), Gen AI Hackathon **Semi-Finalist** GitHub · May 2025

Google ADK, Gemini 2.0 Flash, Vertex AI RAG Engine, PyMuPDF, Python

- Led the team from a system diagram to a working multi-agent assistant that gives non-lawyers a first read of a contract: the clauses that matter and the risks to watch for.
- Built the root agent that validates a Google Drive link and routes the contract to an analyst agent, which extracts the text with PyMuPDF.
- Wrote the corpus-preparation script that loads **510 CUAD v1** contracts into a Vertex AI RAG corpus; the short-contract agent checks each clause against it.

AMA Computer Learning Center E-Library · Lead Programmer & Project Manager, 13-member team 2023

- Built a full-stack e-library with QR-based transactions for the school, deployed on a Linux server (DigitalOcean, PostgreSQL, Nginx) and validated through testing before rollout.
- Led a 13-member team across 3 departments and coordinated with advisers through the rollout; self-financed a year of hosting.

LEADERSHIP & HACKATHONS

- **InnOlympics 2026**, Pangarap sa Kalusugan track: led a 4-person team and owned the system design for Nura, shipped in 24 hours.
- **Gen AI Hackathon, Semi-Finalist** (2025): team lead of Joybread Studios; built JuryBee.
- **MERALCO IDOL 2025, Top 14 Finalist**, selected from a large applicant pool: overall lead, system designer and project manager of a 4-person team that built a civic-reporting platform in one day, with a computer-vision model over user reports, a report-density heatmap and barangay, city and region leaderboards.
- **Trend Micro University Capture the Flag**: ranked 32nd of 200 teams as team JOYBREAD (NU Manila), 285 points.
- **Also competed:** Alibaba Hackathon · HACK4GOV · DLSU Hacker Cup 2025 (MedCheck, multi-agent AI).

ORGANIZATIONS

Google Developer Student Club, NU Manila · Member 2024 – Present

NU Wizards, National University Manila · Competition Committee: organized and ran programming contests, a blockchain seminar and an ideathon 2024 – Present

CERTIFICATIONS

Eskwelabs Innovation Fellowship · Data Modeling Track (Cohort 10) · completed, certificate pending 2026

SKILLS

Machine Learning: scikit-learn, LightGBM, statsmodels, pandas, NumPy, feature engineering, cross-validation, evaluation (ROC-AUC, F2, precision-recall), causal inference, leakage analysis, OpenCV, Tesseract OCR

LLM & Multi-Agent: RAG, pgvector, Vertex AI (RAG Engine, embeddings), Sentence-Transformers, Google ADK, LangChain, Gemini, Groq Llama 3.1, Ollama, prompt design, model fallback

Backend: FastAPI, Django REST Framework (SimpleJWT, drf-spectacular), Spring Boot, Node.js, OpenAPI, JSON Schema

Frontend: React, TypeScript, Next.js, Vite, Tailwind CSS, PWA, Streamlit · **Data:** PostgreSQL, PostGIS, Supabase, Firebase / Firestore

Hardware: ESP8266 (NodeMCU), Arduino C++, SCT-013 current sensors, analog front-end calibration

Tools: Docker Compose, Cloud Run, DigitalOcean, Nginx, n8n, Playwright, pytest, Jupyter, Git and GitHub

Languages: Python, SQL, TypeScript, JavaScript, Java, Kotlin, C++ · **Spoken:** English (fluent), Filipino (native)